

TIM COATES

Search...

[DEVELOPMENT](#), [MENTAL HEALTH](#), [PROGRAMMING](#), [SELF-CARE](#), [SOFTWARE DEVELOPMENT](#),
[TECH SUPPORT](#)

Mental Health of Developers in the Open Source Landscape

Published by **Tim Coates** on April 3, 2024



In the current age, the world runs on code. From websites to mobile apps, software developers are the unseen architects shaping our technological landscape. Within

this realm, open source software (OSS) plays a vital role, collaboratively built and freely available. However, the nature of OSS development presents unique challenges that can significantly impact the mental well-being of these developers.

This article delves into the current state of mental health among software developers, with a particular emphasis on the pressures and potential pitfalls faced by those contributing to open source projects. By understanding these challenges, we can work towards fostering a more supportive environment, ensuring the well-being of coders and the continued success of the open source movement.

Between Then and Now

The original intent of open source was making the source code of software programs and platforms accessible by all. That is, individuals are invited to update the code with the purpose of improving the software for the benefit of all users.

Between then and now, things have changed whereby it can be difficult to contribute to open source projects due to aggressive communications and complex codebases. The author argues that open source should focus on empathy and collaboration again

The Contrasting Realities of Open Source

Open source software offers numerous advantages, fostering innovation, enabling rapid development, and keeping costs down. However, the very aspects that make OSS successful can also contribute to mental health issues for developers.

One significant challenge is the skill mismatch that often occurs when companies task developers with incorporating open source projects that may not necessarily align with their skillset. As Rob Mensching, a software developer, and open source contributor, notes, this mismatch can lead to frustration for both the developer struggling with unfamiliar code and the maintainer reviewing potentially flawed contributions.

Additionally, the pressure to perform can take a toll on developers contributing to popular open source projects. With the larger community scrutinizing their work, many feel immense anxiety and fear of making mistakes, as highlighted by Vadim Kravcenko, a software engineer and mental health advocate.

Furthermore, a defining feature of open source is its reliance on volunteer contributions. While developers gain experience and recognition, the vast majority of maintainers dedicate significant personal time to keeping projects afloat. As Antfu points out, this uncompensated labor can lead to burnout, especially if not balanced with other aspects of life.

Challenges for Developers

The mental health challenges faced by software developers extend beyond the realm of open source. The industry itself is fast-paced and demanding, constantly bombarding developers with new technologies, tight deadlines, and the pressure to deliver flawless code.

Social isolation, loneliness, anxiety and depression are major issues for developers. A cursory glance at posts on Reddit talk to these matters and the effects of spending long hours coding. The advent of remote work can lead to a lack of work-life balance and limited social interaction, which can exacerbate mental health issues.

Imposter syndrome, the feeling of inadequacy despite demonstrable skills, is also prevalent in the developer community, leading to anxiety and self-doubt. As one anonymous developer shared, “No matter how much I accomplish, I can’t shake the fear that I’m not good enough.”

Moreover, the constant pressure to perform can contribute to burnout, characterized by emotional exhaustion, cynicism, and a reduced sense of accomplishment. This can have severe consequences for both the individual’s well-being and the quality of their work.

Support for Developers

Fortunately, there are steps that can be taken to create a more supportive environment for software developers, both within and outside the open source realm.

Open source adoption can benefit businesses of all sizes – larger companies can invest in training programs to empower developers, while smaller businesses can leverage managed open-source services or well-documented libraries. Both large and small businesses should prioritize fostering a collaborative environment where developers can share knowledge and overcome challenges together. This can be achieved through internal knowledge-sharing sessions or company-backed documentation platforms.

Raising awareness about mental health issues in the tech industry and normalizing seeking help is crucial. As SurveyPoint.ai emphasizes, companies can provide resources like access to therapy or Employee Assistance Programs (EAPs) to support their employees' mental well-being.

Within open source communities, creating a culture of appreciation for contributors can go a long way. Acknowledging their efforts and providing positive reinforcement can alleviate the pressure and anxiety that often accompany contributing to these projects.

Finally, promoting sustainable practices, such as encouraging healthy work-life balance and setting realistic deadlines, can prevent burnout among both employed and volunteer developers.

Establishing clear contribution guidelines and expectations can also help alleviate stress for developers approaching open source projects. As Mensching suggests, well-defined processes can streamline interactions and reduce potential conflicts or misunderstandings.

Final Thoughts

The mental health of software developers is paramount to the continued success of the entire tech industry, especially in the open source domain. By acknowledging

the challenges they face and implementing supportive measures, we can ensure a thriving developer community that is both creative and well-balanced.

And regardless of your role, it's essential to prioritize self-care and seek help when needed – healthy developers write healthy code, and a healthy codebase is the foundation for a healthy digital world.

Remember, addressing mental health issues in the software development industry is not just a matter of individual well-being; it's a collective responsibility that will shape the future of technology and the world we live in.

References

1. "Open Source and Mental Health: The Biggest Challenges Facing Developers." Information Age, <https://www.information-age.com/open-source-mental-health-biggest-challenges-facing-developers-19370/>.
2. Dignan, Larry. "5 Ways to Improve Mental Health for Software Developers." TechCrunch, 11 Nov. 2021, <https://techcrunch.com/2021/11/11/5-ways-to-improve-mental-health-for-software-developers/>.
3. "Access to Mental Health Support in the Digital Age." SurveyPoint.ai, 9 Nov. 2023, <https://surveypoint.ai/blog/2023/11/09/access-to-mental-health-support-in-the-digital-age/>.
4. Kravcenko, Vadim. "Mental Health in Software Engineering." Vadim Kravcenko, <https://vadimkravcenko.com/shorts/mental-health-in-software-engineering/>.
5. Antfu. "Mental Health in OSS." Antfu's Blog, 14 Aug. 2022, <https://antfu.me/posts/mental-health-oss>.
6. Mensching, Rob. "A Microcosm of the Interactions in Open Source Projects." Rob Mensching's Blog, 30 Mar. 2024, <https://robmensching.com/blog/posts/2024/03/30/a-microcosm-of-the-interactions-in-open-source-projects/>.
7. "Open Source." Investopedia, <https://www.investopedia.com/terms/o/open-source.asp#:~:text=feedback%20and%20improvements-,The%20Bottom%20Line,accessible%20by%20only%20its%20creator>.
8. Heilmann, Christian. "A Worrying Change in Open Source Perception." Christian Heilmann's Blog, 11 July 2019, <https://christianheilmann.com/2019/07/11/a-worrying-change-in-open-source-perception/>.

7 responses to “Mental Health of Developers in the Open Source Landscape”



Workers' Archive

April 5, 2024

Hi Tim, interesting piece. I wondered what “company-backed documentation platforms” are? I don’t have any experience of coding but have worked with developer colleagues. Does much horizontal-type teamwork occur? Eg. people on the same level working together, as opposed to on their own or within a hierarchical set-up?

PS. Would it be possible to share this post on my blog with a link and credit to you and your blog? I share stories connecting work and mental health.

★ Like

Reply



Tim Coates

April 5, 2024

Firstly, please share.

★ Like

Reply



Tim Coates

April 5, 2024

Secondly ... company backed documentation platforms are tools like Wiki or SharePoint or similar where knowledge can be shared. When supported by the company there can be a incentive to use it, or less likely to be told it's a waste of time.

Open source development is Horizontal teamwork. So is "pair programming". it harder where there are limited resources (programmers), or when there are time pressures to deliver.

★ Like

Reply



Workers' Archive

April 5, 2024

Thanks for explaining. I was also curious about when companies ask staff to incorporate open source projects? Is it to integrate with their software? And, also, why would there be a skills mismatch – does open source tech. involve different types of coding? Hope it's ok to ask these questions. I'll incorporate them into my post when I share.

★ Like



Tim Coates

April 6, 2024

Why use Open Source?

- Cost saving is an obvious benefit
- Not locked into a particular propriety solution/vendor
- Popularity of a particular open source product

- Get or access source code to product
- Can change source code if needed
- Increased flexibility

Is it to integrate with their software?

There are many way you can look at integration and will conditionally say YES.

The easiest example would be in software development and bug tracking software. (Going back many years, Microsoft had own version control (Source Safe), but it was a pain. Secondly, only supported Windows. Around 2000, I was working with a group that targeted Linux, Windows, and something-else and needed a solution supporting each. The best here was found in the open source space.)

Back to the bug tracking ... Find the product, download and then install it. Works fine. Sales people (?) can enter bugs into the system if needed. Cool. And then some error occurs (it's a bug?) – missing file, or incorrect formatting of date, or even missing data? The boss asks you to fix the problem. Your skill set is in C and C++, but the bug tracking system was developed using php. You may have a basic understanding and have written a couple of scripts in php. You are now moving into a space that you might not have worked before – it web based, source code base much larger, and your understanding of the system code-wise and frameworks used in minimal. This is the skill mismatch. This is just one example. There might also be admin related mismatches and other areas.

Does open source tech. involve different types of coding?

As in the above example... very possible or Yes (from the above example)

Hope it's ok to ask these questions ...

Of course. I don't have any problems with any questions you might want to throw my way. Keem 'em coming.

★ Like



Also, I wondered how typical it is for developers to work with open source coding? Is it a kind of voluntary training expected in the industry?

★ Like

Reply



Tim Coates

April 5, 2024

the cynical side of me says that it can make you look better if you (can) put this on your CV.

Another way might be ... you have just installed package X, and there is some bug in the code you find. You fix it (because you can) and can then get it added to repository of the code base. (talking high level here)

Finally, some might volunteer time with a NFP, and someone else might choose to spend time on open source – in the latter option you need to have a passion for this.

★ Liked by 1 person

Reply

Hello,

I'm Tim



I'm a (former) software developer, occasional coder, and mental health advocate. I write about coding using modern Pascal (fpc, Delphi, etc.) and mental well-being—two very different worlds, and both run better with a little care, attention, and creativity.

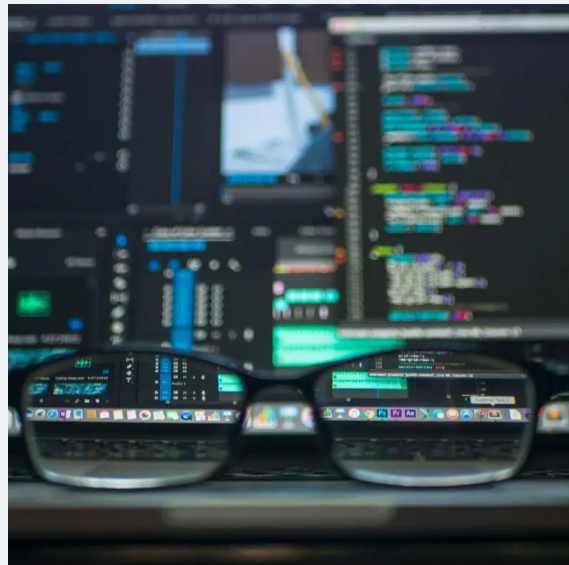
Recent posts



Programmer Mental Health:
Celebrating Small Wins

Advanced Records: A Comparison
with Pascal Records and Object-
Oriented Classes

(Why) Pascal Deserves a
Second Look



LazLogger: A Lightweight Logging
Solution in Lazarus

```
a type
am Project2;

TYPE CONSOLE}

.res}

tem.SysUtils, classes;

tFile = 'c:\temp\textfile.txt';

ion CheckIOResult: boolean;

  LResult := IOResult;
  LResult <> 0 then
in
```

Parts of a Pascal Program

Programming as a Creative Craft

TIM COATES

Designed with [WordPress](#)